

Low-Cost Acoustic Array for Small UAV Detection and Tracking

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Abstract—This paper describes a low-cost acoustic array designed to locate and track small unmanned aerial vehicles (UAVs). The goal of this project was to create an inexpensive array with dynamically placed microphones that could detect and locate broadband sources in the far-field through algorithms for array calibration and beamforming. Experiments performed using remote controlled airplanes in an open field demonstrated favorable results.

I. INTRODUCTION

Unmanned aerial vehicles currently pose a disproportionate threat to ground forces due to the inability of most sensors to detect them. UAVs, as small, low-velocity aircraft, present weak signatures to RF, EO, and IR systems. As such, standard air defense is ill-prepared for UAV detection [1]. An acoustic array, however, does not depend on the size of the aircraft for detection, but rather the sound of the engine, and can therefore serve as an effective means of detecting and tracking many unmanned aerial vehicles.

The goal of this project was to design an acoustic array using commercial off-the-shelf (COTS) hardware that would be mobile and that could adapt to different signatures and deployment environments. Since beam-width is dependent on the target frequency and the distance between microphone elements, the positions of the elements needed to be variable, thus requiring a calibration procedure to determine the locations of the microphones each time they are placed. Software was also developed to perform signal preconditioning and beamforming.

Some arrays consist of fixed elements to eliminate the need for calibration, while various techniques have been used to determine microphone locations for dynamic arrays. One such technique includes the use of global-positioning systems on each microphone. Other methods use known source locations to interpolate the microphone locations [2], while others with unknown source locations use the weights from the eigenvector associated with the largest eigenvalue of the cross-correlation matrix [3]. Various source location methods include: generalized cross-correlation, least-squares approximation, and maximum likelihood estimation [2].

The array discussed in this paper is built to be both low-cost and mobile while retaining the ability to detect and track UAVs. It consists of 24 microphones, where the locations of the microphones are initially unknown, and uses a delay-and-sum beamforming algorithm.

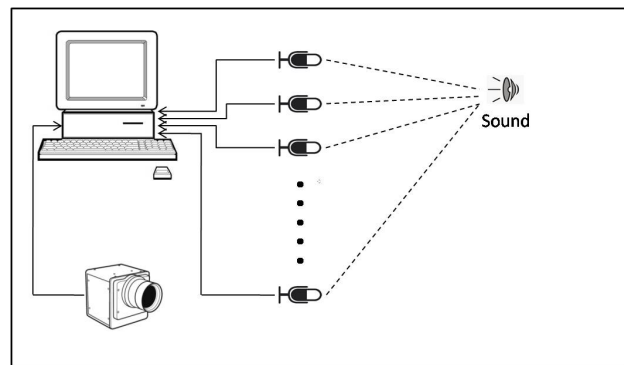


Fig. 1. Hardware setup showing a linear microphone array with an IR camera.

The remainder of this paper is organized as follows. Section II discusses the acoustic array, hardware, and the software developed for signal processing. The components of the array are described first, followed by a description of the test setup and data collection process. We then detail the calibration procedure necessary to determine the locations of the microphones in the test space and the beamforming algorithm used for detection and tracking. Section III presents the results of the acoustic array when tested with a UAV in an open field. Finally, Section IV presents our conclusions regarding the array and its performance.

II. SYSTEM DESIGN

A. Hardware

The hardware used for this project consists of twenty-four custom-made microphones, a COTS sound-recording board, a desktop Macintosh G4, and a video camera (used as a visual reference). The microphone design comprises a condenser microphone cartridge and a MOSFET for pre-amplification (Figure 2). Custom building the microphones allowed a high-quality Panasonic microphone cartridge and amplifier circuit to be assembled at a fraction of the cost of a commercial microphone. The microphone cartridge and pre-amp were enclosed in a Fisherman's Friend throat lozenge box (similar to an Altoid case) and powered by a 9-volt battery. Ports are drilled in the metal casing for the microphone cartridge, a power switch, and a twenty-foot output cable with a quarter-inch jack. Each microphone cost \$14.50 a piece to construct.

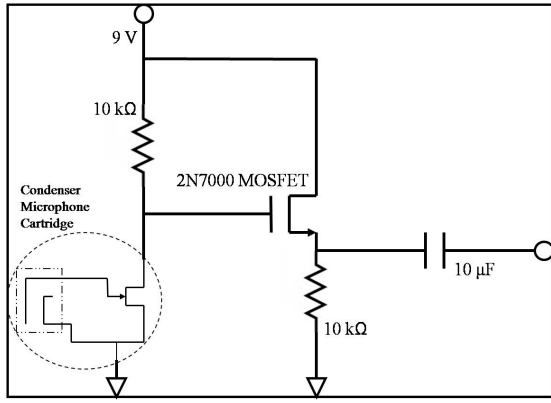


Fig. 2. The above is a diagram showing the design used for the custom-built microphones. The circuit components shown in the dotted lines represents the internal components of the Panasonic microphone cartridge. The circuit connected to the microphone cartridge represents the pre-amp design.

To interface between the microphones and the computer, a sound recording board by Mark of the Unicorn (MOTU) was used. This sound recording board has 24 I/O channels with a sampling frequency of 44.1kHz and a maximum bandwidth of 96kHz per channel. A basic IR camera was used as a point of reference for calibration. All signal processing is performed using MATLAB. Acoustic data from each microphone is saved as a separate audio file resulting in up to 24 audio channels per data collection. With costs being \$1,420 for the MOTU board, \$2,000 for a refurbished Macintosh G4, and \$348 for 24 microphones, the total cost of an acoustic array of this design is \$3768.

B. Array Configuration

The above described hardware supports an array of up to $M = 24$ microphones, though different numbers of microphones were used in subsequent experiments. Initially, the microphones are placed in a nominally straight line on the ground with a spacing of approximately 10 inches between elements. The 2-dimensional locations of the microphones (x_m, y_m) for $m = 1, 2, \dots, M$ are assumed to be unknown. Audio data collected by each microphone is represented as

$$x_m(t) = s(t - t_m) + n_m(t) \quad m = 1, \dots, M, \quad (1)$$

where $s(t)$ is the target audio signal and $n_m(t)$ is unknown noise and clutter experienced by microphone m . The time delay from the target to the m th microphone is t_m . A target in the far-field at bearing ϕ has time delay $t_m = x_m \cos \phi + y_m \sin \phi$, where $\phi = 0^\circ$ is broadside to the array.

When the microphones are in place, broadband calibration signals, in this case chirps generated in MATLAB, are emitted periodically within the field of regard of the array. Any number of chirps can be used, but more chirps result in more accurate calibration. All chirps are generated within one data collection so that the calibration data consists of M audio tracks with C chirps per audio track.

C. Calibration

Calibration is needed to determine the x, y -locations of the microphones relative to the target space. This is done by using the $M \times C$ time delay of arrival (TDOA) matrix associated with the microphones and corresponding chirps to interpolate the microphone locations. This matrix consists of time delays, $t_{m,c}$, representing the number of samples (at 44.1kHz) that occur between when chirp c occurs and when the same chirp reaches microphone m .

$$TDOA = \begin{bmatrix} t_{1,1} & t_{1,2} & \dots & t_{M,1} \\ t_{2,1} & t_{2,2} & \dots & t_{M,2} \\ \vdots & \vdots & \ddots & \vdots \\ t_{1,C} & t_{2,C} & \dots & t_{M,C} \end{bmatrix} \quad c = 1, 2, \dots, C$$

The TDOA matrix is found via the cross-correlation of each microphone channel, $x_m(t)$, and the MATLAB generated reference chirp, $r(t)$. Because the chirp is created in MATLAB, it can be exactly replicated digitally. The samples at which the recorded channels are most highly correlated with the reference signal indicate the relative time delays of arrival:

$$t_{m,c} = \arg \max_{\tau} \left[\sum_t h(t - t_c) x_m(t) r(t - \tau) \right]. \quad (2)$$

The c th chirp is selected by the windowing function $h(t)$. The resultant matrix of time delays of arrival is normalized by identifying the center microphone as having zero time delay and relating the other TDOAs respectively, resulting in both positive and negative TDOAs.

The TDOA matrix is smoothed using a polynomial fit along each row to remove outlying time delay of arrival errors. The x and y coordinates of the microphones relative to the test area are then found using an unconstrained nonlinear optimization of the smoothed TDOA matrix.

Calibration need only be performed once for each data collection so long as the microphones remain in the same location. Microphone locations are saved, allowing a user to leave the array in one position for numerous data collections.

D. Preconditioning

After calibration, data collection begins, and the data is, once again, saved as M audio signals. Before beamforming, some preconditioning of the recorded signals is required in order to suppress interference and ambient noise clutter. For the experiment outlined in Section III, a band-pass filter from 450Hz to 3kHz was applied to each channel of data. A 450Hz cutoff is low enough to remove most of the ambient clutter noise. A 3kHz cutoff was selected to allow the recorded data to be downsampled from 44.1kHz to roughly 8kHz, thus greatly increasing the processing throughput without loss of significant signal energy. This preconditioning allows for beamforming to be performed more efficiently.

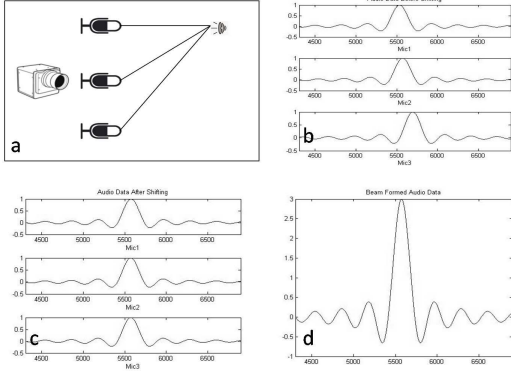


Fig. 3. The diagram above demonstrates the delay-and-sum concept for three microphones. A shows an audio pulse arriving at each microphone. B demonstrates that the resulting pulse is shifted in time for each microphone. C shows the pulses after they have been shifted by the TDOAs and D demonstrates that the resulting delay-and-summed pulse has three times the amplitude of each individual pulse.

E. Beamforming

Once the calibration is complete and each channel has been filtered and downsampled, delay-and-sum beam forming is straightforward. For a particular angle, ϕ , the time delay of arrival for each data channel is computed as

$$t_m(\phi) = x_m \cos \phi + y_m \sin \phi \quad (3)$$

The beamformed signal at aspect ϕ is thus

$$X_B(t; \phi) = \sum_{m=1}^M x_m(t - t_m(\phi)) \quad [3] \quad (4)$$

Non-coherent integration is calculated over segments of $\delta = 1/30$ seconds for each beam to yield the waterfall image

$$E_B(\tau, \phi) = \int_{\tau}^{\tau+\delta} |X_B(t; \phi)|^2 dt. \quad (5)$$

The result is plotted for the duration of the data collection versus DOA(ϕ) and time(τ).

III. RESULTS

The array was tested on remote controlled aircraft that flew with varying flight paths in the far-field (relative to the array). The goal of the experiment was to determine the ability of the array to detect and track UAVs. The array was placed linearly in an open, outdoor field.

After calibration, data collections each consist of a single UAV flight path in the three-dimensional space in front of the linear array. Beamforming was performed in 1° increments over a field of view of -45° to 45° , where 0° is perpendicular to the array.

For this experiment, beamforming was applied to four flight trajectories: a straight path towards the array (Figure 4), a zig-zag pattern that crossed out of the field of view (Figure 5), a larger zig-zag pattern in the field of view (Figure 6), and a circular flight path (Figure 7). A waterfall plot was generated for each flight path and are shown adjacent to the

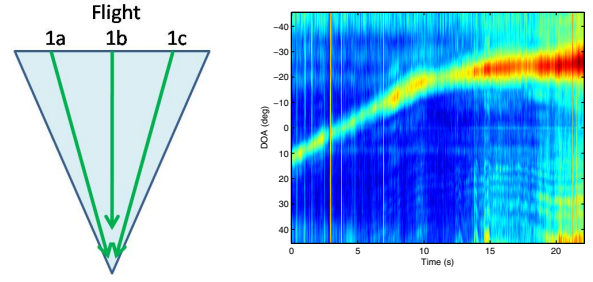


Fig. 4. Tracking flight path 1: straight path towards the array.

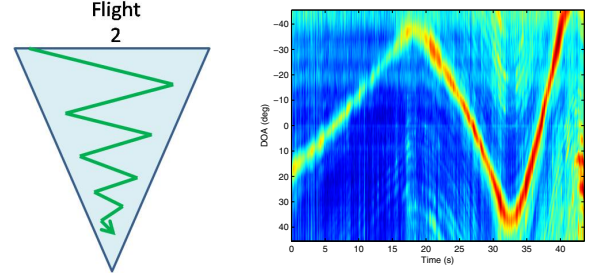


Fig. 5. Tracking flight path 2: Zig-zag that passed out of the field of view.

corresponding executed flight trajectories.

In each plot, one can observe the presence of grating lobes due to the aliasing of high frequency signal content. This is particularly noticeable for flights with the UAV closer to the array and with a higher SNR. With 10-inch element spacing, acoustic signals with wavelengths shorter than 20-inches or frequencies larger than roughly 700Hz will experience spatial aliasing. Several low-level interference sources also appear consistently at aspects of -40° , -30° , -20° , and 10° for example. There are also short-time events in the near field, such as roughly 3 second into flight 1a. In general, the beamformed results and waterfall plots are consistent with the actual flight paths.

IV. CONCLUSION

This acoustic array shows promising results while still fulfilling the initial goals of the project, which were for the array to be both low-cost and mobile. The hardware consists of hand-made and commercial off-the shelf equipment. The

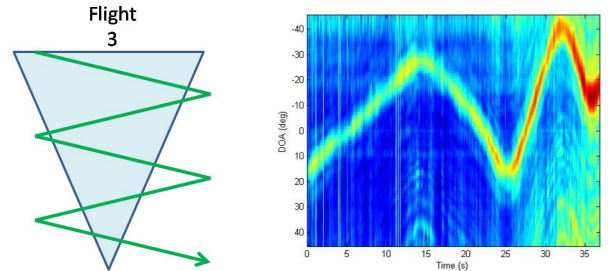


Fig. 6. Tracking flight path 3: Zig-zag in the field of view.

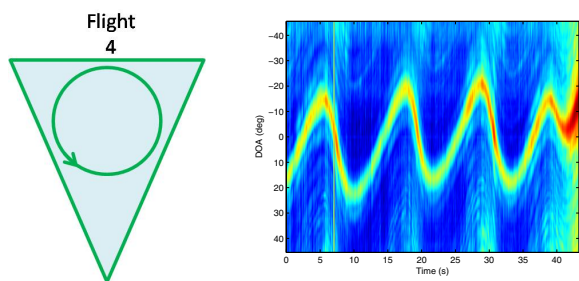


Fig. 7. Tracking flight path 4: three circles in front of the array. Note that because the array is linear, the circular flight path looks sinusoidal. The image on the left is the planned trajectory and the image on the right is the tracked trajectory.

signal processing software is relatively simple and easy to implement, consisting only of a calibration procedure and beamforming algorithm. Future work will be performed to eliminate the calibration procedure and to perfect the post-beamforming result using matched filtering.

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